

BAULKHAM HILLS HIGH SCHOOL

YEARLY EXAMINATION

2009

YEAR 11

**MATHEMATICS
ADVANCED**

Time Allowed: Three Hours

Directions to Candidates:

- Approved calculators may be used
- Begin each question on a new page
- Write your name and your teacher's name at the top of each page
- Show all necessary working

Question 1 (12 marks) Start on a SEPARATE page

- a) Evaluate correct to 3 significant figures

2

$$\sqrt{\frac{39.4 - 13.55}{8.63^2 - 1.98}}$$

- b) Solve
- $\frac{x+5}{3} - 2x = 7$

2

- c) Simplify
- $2\sqrt{2} + \sqrt{18}$

2

- d) Express
- $0.\dot{2}\dot{4}$
- as a fraction in its simplest form

2

- e) Find the exact value of

i) $\sin 60^\circ$

1

ii) $\cos 225^\circ$

1

- f) Factorise and simplify

2

$$\frac{x^2 - 5x - 24}{x^2 + 3x}$$

Question 2 (12 marks) Start on a SEPARATE page

a) Differentiate

i) $y = 6x^2 - 8x + 7$ 1

ii) $y = 2\sqrt{x} - \frac{4}{x}$ 2

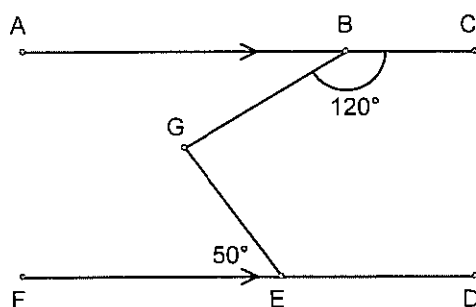
iii) $y = \frac{5x^2}{2x+7}$ 2

b) Solve simultaneously

$$\begin{aligned} 4x - y + 3 &= 0 \\ xy &= 1 \end{aligned}$$
3

c) If $\frac{1}{3+2\sqrt{2}} = a + b\sqrt{2}$ find a and b 2

d) Find the value of the reflex angle $\angle BGE$ 2
Give reasons.



Question 3 (12 marks) Start on a SEPARATE page

a) Solve $x^2 - 6x + 5 \geq 0$ 2

b) If $f(x) = \begin{cases} -4 & \text{for } x \leq -2 \\ x & \text{for } -2 < x < 1 \\ x^2 - 1 & \text{for } x \geq 1 \end{cases}$

i) Find $f(-1)$ 1

ii) Find $f(a^2 + 1)$ 2
Simplify your answer.

iii) Sketch the graph of $f(x) = y$. 3

c) Given $\sin \alpha = -\frac{3}{7}$ and $\tan \alpha > 0$, find the exact value of $\sec \alpha$ 2

d) Solve for $0^\circ \leq \theta \leq 360^\circ$ 2
$$\sin \theta + \frac{\sqrt{3}}{2} = 0$$

Question 4 (10 marks) Start on a SEPARATE page

a) i) Show that $\frac{x}{x+2} = 1 - \frac{2}{x+2}$

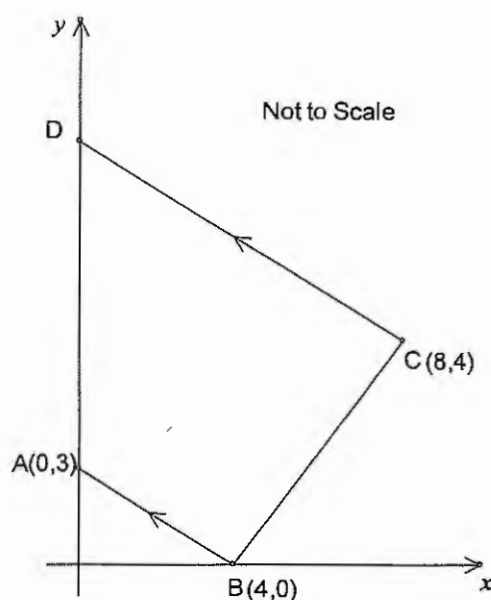
1

ii) Hence or otherwise, sketch the following graph showing all important features

$$y = \frac{x}{x+2}$$

2

b)



Given $A(0,3)$, $B(4,0)$, $C(8,4)$ and $AB \parallel CD$

i) Find the gradient of AB

1

ii) Show that the equation of AB is $3x + 4y - 12 = 0$

2

iii) Find the coordinates of D

1

iv) Find the distance of AB

1

v) Find the perpendicular distance of C from AB

2

vi) Find the area of $ABCD$. Answer in exact form.

2

Question 5 (10 marks) Start on a SEPARATE page

a) Solve for $0^\circ \leq \theta \leq 360^\circ$

i) $\cos^2 \theta - \frac{1}{2} = 0$ 2

ii) $\sin \theta = \sqrt{3} \cos \theta$ 2

b) Solve $|4x - 2| \geq 14$. Sketch your solution on the number line 3

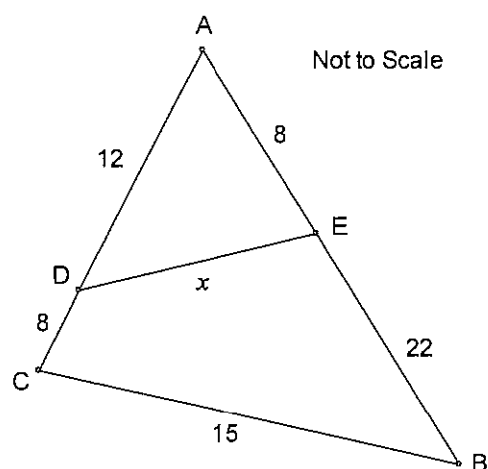
c) The point $P(x, y)$ moves such that PA is perpendicular to PB . 3

Find the locus of P if A is the point $(-3, 6)$ and B is the point $(-7, -6)$

Describe the locus.

Question 6 (13 marks) Start on a SEPARATE page

- a) Given that $AD = 12$, $AE = 8$, $DC = 8$, $CB = 15$ and $EB = 22$



- i) Prove that $\triangle AED \parallel \triangle ACB$

3

- ii) Hence, find the value of x , justifying your working.

2

- b) Evaluate $\sum_{n=1}^{\infty} 3^{-n}$

2

- c) For what values of k does the quadratic equation $x^2 - (k+3)x + 4k = 0$

- i) have no real roots.

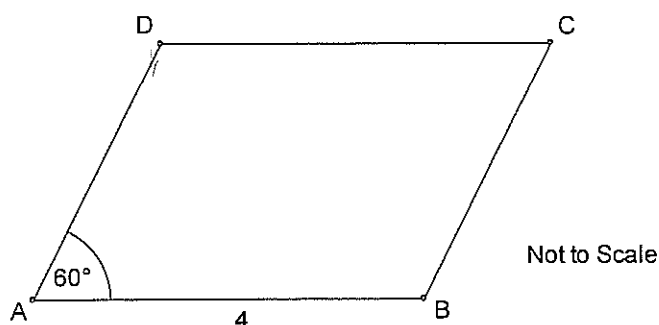
3

- ii) have one real root equal to -3.

1

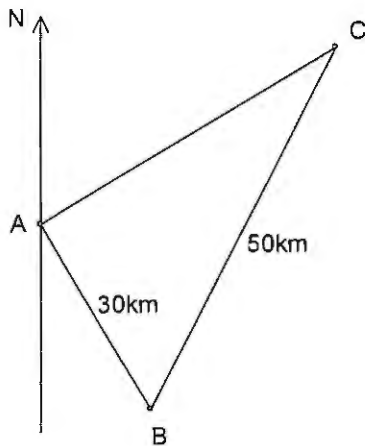
- d) Find the area of a rhombus $ABCD$ with sides of 4cm and $\angle DAB = 60^\circ$
Answer as an exact value.

2



Question 7 (10 marks) Start on a SEPARATE page

- a) Two friends drive 30km from A on a bearing of 150°T to point B . They then drive from B to C , a further 50km on a bearing of 050°T .



- | | |
|--|---|
| i) Show that $\angle ABC = 80^\circ$ | 1 |
| ii) Find the distance from A to C | 2 |
| iii) Find the bearing of A from C
Show all the working | 2 |
| | |
| b) Determine if the function $f(x) = \frac{1}{(x-1)(x+1)}$ is an odd or even function, or neither.
Justify your answer. | 2 |
| | |
| c) Given $kx^2 - (k+1)x - 2k + 3$
find for what value(s) of k , the expression above is negative definite. | 3 |

Question 8 (12 marks) Start on a SEPARATE page

-
- | | | |
|------|---|----------|
| a) | The first three terms of a geometric series are $2x + 2$, $5x + 1$ and $10x + 2$. Find the value(s) of x . | 2 |
| b) | Simplify $\sec(180^\circ - \theta) \cdot \tan(90^\circ - \theta)$ | 3 |
| c) | On a number plane, shade the region satisfying both the inequalities
$y > \sqrt{9 - x^2}$ and $y \geq x^2 + 3$ | 3 |
| d) | The quadratic equation $2x^2 - 3x + 6 = 0$ has roots α and β .
Find the value of | |
| i) | $\alpha + \beta$ | 1 |
| ii) | $\alpha \beta$ | 1 |
| iii) | $\alpha^3 + \beta^3$ | 3 |

Question 9 (12 marks) Start on a SEPARATE page

a) Given $y = x^3 - 2x^2 + x$ i) Find $\frac{dy}{dx}$ 1

ii) Find the stationary point(s) and determine their nature 3

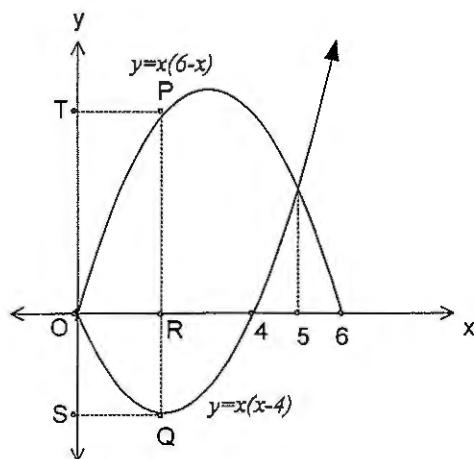
iii) Find the point(s) of inflexion.
Justify your answer. 2iv) Sketch this curve for $-1 \leq x \leq \sqrt{2}$ showing all relevant features 2v) State where this function is increasing **and** concave up. 1b) Prove $(\cot \theta + \operatorname{cosec} \theta)^2 = \frac{1 + \cos \theta}{1 - \cos \theta}$ 3

Question 10 (10 marks) Start on a SEPARATE page

- a) In the diagram below, P is a point on the curve $y = x(6 - x)$ and Q is a point on the curve $y = x(x - 4)$

PQ cuts the x -axis at right angles at R .

S and T are points on y -axis such that $PQST$ is a rectangle.



- i) Show that the length of PQ is given by $10x - 2x^2$ 2
 - ii) Find an expression for the area $PQST$ as a function of x 1
 - iii) Find the value of x which gives the maximum area for $PQST$ ($0 \leq x \leq 5$) 2
Justify your answer.
- b) Given that the line $lx + my + n = 0$ is a tangent to $x^2 = 4ay$, 3
Prove that $a = \frac{mn}{l^2}$.

End of Paper

Yr. 11 - yearly 2009 - Zunit

AnswersQuestion 1 ⑫

a) $0.59713... = 0.597$

b) $\frac{x+5}{3} \Rightarrow 2x = 7$

$x+5 - 6x = 21$ ✓

$-5x = 16$

$x = -16/5$ ✓

c) $2\sqrt{2} + \sqrt{18} = 2\sqrt{2} + \sqrt{9 \times 2}$
 $= 2\sqrt{2} + 3\sqrt{2} = 5\sqrt{2}$

d) $x = 0.2424...$
 $100x = 24.2424...$ ✓
 $99x = 24$
 $x = \frac{24}{99}$ ✓

e) i) $\sin 60^\circ = \frac{\sqrt{3}}{2}$ ✓

ii) $\cos 225^\circ = -\frac{1}{\sqrt{2}}$ ✓

f) $\frac{x^2 - 5x - 24}{x^2 + 3x} = \frac{(x+3)(x-8)}{x(x+3)}$ ✓
 $= \frac{x-8}{x}$ ✓

Question 2 ⑫

a) i) $\frac{dy}{dx} = 12x - 8$ ✓

ii) $y = 2x^{\frac{1}{2}} - 4x^{-1}$
 $\frac{dy}{dx} = 1x^{-\frac{1}{2}} + 4x^{-2}$ ✓

iii) $\frac{dy}{dx} = \frac{10x(2x+7) - 2 \times 5x^2}{(2x+7)^2}$ ✓
 $= \frac{20x^2 + 70x - 10x^2}{(2x+7)^2}$
 $= \frac{10x^2 + 70x}{(2x+7)^2}$ ✓

b) (1) $4x - y + 3 = 0$
(2) $xy = 1$
(2) $y = \frac{1}{x}$ ✓

(1) $4x - \frac{1}{x} + 3 = 0$
 $4x^2 - 1 + 3x = 0$
 $x = \frac{-3 \pm \sqrt{9+16}}{8}$

$(4x-1)(x+1) = 0$

$\boxed{x = 1/4}$
 $\boxed{y = 4}$ ✓

$\boxed{x = -1}$
 $\boxed{y = -1}$ ✓

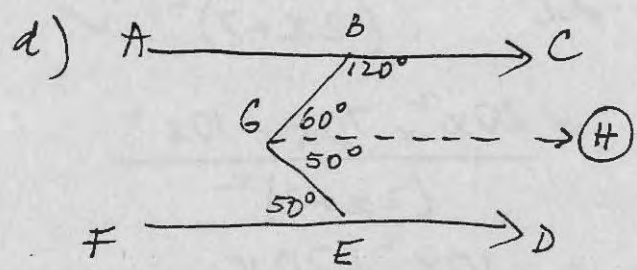
Question 2-cont.

c) $\frac{1}{3+2\sqrt{2}} = a+b\sqrt{2}$

$\frac{3-2\sqrt{2}}{9-4 \times 2} = a+b\sqrt{2}$

$3-2\sqrt{2} = a+b\sqrt{2}$

$a=3, b=-2$



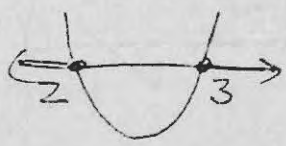
$\angle BGH = 60^\circ$ CO-int. $\angle$'s on $AC \parallel GH$ are suppl.

$\angle HGE = 50^\circ$ Alternate $\angle$'s on $GH \parallel FE$

$\therefore \angle BGE = 360^\circ - 60^\circ - 50^\circ = 250^\circ$ (angles at point = 360°)

Question 3

a) $x^2 - 6x + 5 \geq 0$
 $(x-2)(x-3) \geq 0$

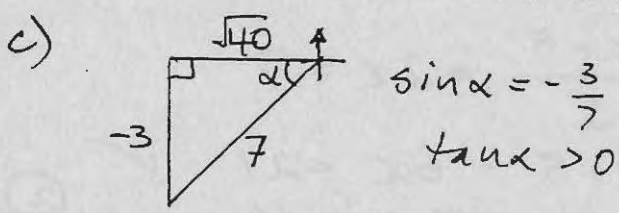


$x \leq 2$ or $x \geq 3$

b) i) $f(-1) = -4$ ✓
 ii) $f(a^2+1) = (a^2+1)^2 - 1$ ✓

$= a^4 + 2a^2 + 1 - 1 = a^4 + 2a^2$

(iii) See at end of solns. ✓✓✓



$\sin \alpha = -\frac{3}{7}$
 $\tan \alpha > 0$

$\sec \alpha = \frac{1}{\cos \alpha} = \frac{1}{-\frac{\sqrt{40}}{7}} = -\frac{7}{\sqrt{40}}$

d) $\sin \theta + \frac{\sqrt{3}}{2} = 0$

$\sin \theta = -\frac{\sqrt{3}}{2}$ $\theta = 60^\circ$ acute

$\therefore \theta = 240^\circ, 300^\circ$

Question 4

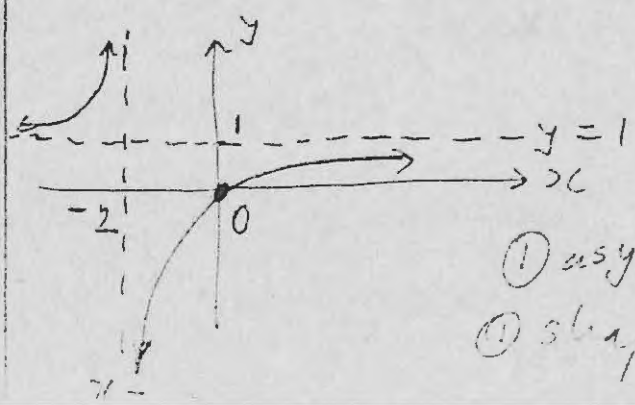
(12)

a) i) $\frac{x}{x+2} = 1 - \frac{2}{x+2}$

RHS = $1 - \frac{2}{x+2} = \frac{x+2-2}{x+2}$

$= \frac{x}{x+2} = \text{LHS}$

ii) $y = \frac{x}{x+2} = 1 - \frac{2}{x+2}$



① asymptotes
 ② shape

Question 4b

i) $m_{AB} = -\frac{3}{4}$ ✓ $A(0,3)$
 $B(4,0)$

ii) $y - 3 = -\frac{3}{4}(x - 0)$ ✓

$$y = -\frac{3}{4}x + 3$$

∴ $\frac{3}{4}x + y - 3 = 0$ ✓
 $3x + 4y - 12 = 0$

iii) $D(0,10)$ ✓

iv) $AB = \sqrt{4^2 + 3^2} = 5$ ✓

v) $C(8,4)$ $3x + 4y - 12 = 0$
 $d = \frac{|3 \times 8 + 4 \times 4 - 12|}{5} = \frac{28}{5}$ ✓

vi) $AB = 5$

$$CD = \sqrt{8^2 + 6^2} = 10$$

∴ Area = $\frac{1}{2} \times \frac{28}{5} (5 + 10) = \frac{14}{5} \times 15$
 $= 42$ ✓

Question 5

(10)

a) i) $\cos \theta = \frac{1}{2}$

$$\cos \theta = \frac{1}{2}$$

$$\cos \theta = \pm \frac{1}{\sqrt{2}}$$

$\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$
✓ ✓

ii) $\sin \theta = \sqrt{3} \cos \theta$

$$\frac{\sin \theta}{\cos \theta} = \sqrt{3} = \tan \theta$$

∴ $\theta = 60^\circ, 240^\circ$
✓ ✓

b) $|4x - 2| \geq 14$

$$4x - 2 \geq 14$$

$$4x \geq 16$$

$$x \geq 4$$

$$4x - 2 \leq -14$$

$$4x \leq -12$$

$$x \leq -3$$

OR



$x \leq -3$, $x \geq 4$
✓ ✓

c) $m_{PA} \times m_{PB} = -1$

$A(-3,6)$ $B(-7,-6)$ $P(x,y)$

$$\frac{y-6}{x+3} \times \frac{y+6}{x+7} = -1$$
 ✓

$$y^2 - 36 = -(x^2 + 10x + 21)$$

$$x^2 + 10x + y^2 = -21 + 36$$
 ✓

$$(x+5)^2 + y^2 = -21 + 36 + 25$$

$$(x+5)^2 + y^2 = 40$$

∴ circle centre $(-5,0)$

$$r = \sqrt{40}$$
 ✓

Question 6 (13)

In $\triangle AED$ and $\triangle ACB$:-

a) $\angle A$ is in common

$$\frac{AE}{AC} = \frac{8}{20} = \frac{2}{5} \quad \checkmark$$

$$\frac{AD}{AB} = \frac{12}{30} = \frac{2}{5} \quad \checkmark$$

$\therefore \triangle AED \sim \triangle ACB$

(two matching sides are in the same ratio and the included $\angle$'s =).

$$\text{ii) } \frac{x}{15} = \frac{2}{5} \therefore x = 6 \quad \checkmark$$

(since matching sides in similar triangles are in same ratio)

$$\text{b) } \sum_{n=1}^{\infty} 3^{-n} = \frac{1}{3} + \frac{1}{9} + \dots \quad a = \frac{1}{3} \quad r = \frac{1}{3} \quad \checkmark$$

$\therefore$ sum exists $\because |r| < 1$

$$\therefore \sum_{n=1}^{\infty} 3^{-n} = \frac{\frac{1}{3}}{1 - \frac{1}{3}} = \frac{\frac{1}{3}}{\frac{2}{3}} = \frac{1}{2} \quad \checkmark$$

$$\text{c) } x^2 - (k+3)x + 4k = 0$$

$$a = 1 \quad b = -(k+3) \quad c = 4k$$

(i) $\Delta < 0$ $\checkmark$ no real roots

$$b^2 - 4ac = [-(k+3)]^2 - 4 \times 1 \times 4k < 0$$

$$\therefore k^2 + 6k + 9 - 16k < 0$$

$$k^2 - 10k + 9 < 0$$

$$(k-9)(k-1) < 0 \quad \checkmark$$

$$\frac{1}{1} \quad \frac{1}{9}$$

$$1 < k < 9 \quad \checkmark$$

ii) if $x = -3$

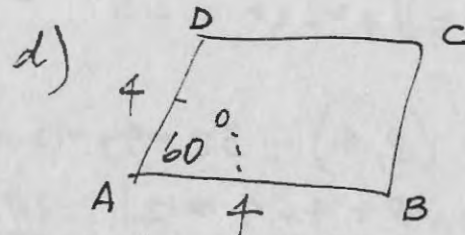
$$\therefore x^2 - (k+3)x + 4k = 0$$

$$(-3)^2 - (k+3)x - 3 + 4k = 0$$

$$9 + 3k + 9 + 4k = 0$$

$$7k = -18$$

$$\therefore k = -18/7 \quad \checkmark$$



$$A = \left(\frac{1}{2} \times 4 \times 4 \times \sin 60^\circ \right) \times 2$$

$$= 16 \frac{\sqrt{3}}{2} \quad \checkmark$$

$$\therefore A = 8\sqrt{3} \text{ cm}^2$$

Question 7



i) $\angle SAB = 30^\circ$

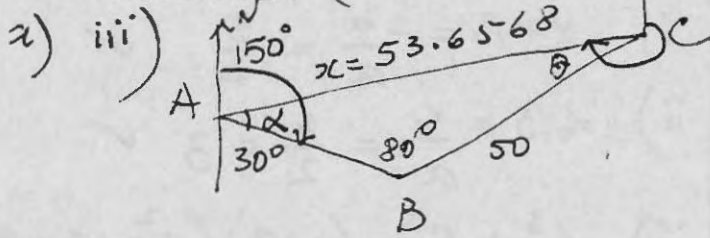
$\angle ABN' = 30^\circ$ alt. $\angle$'s on parallel lines
 $\therefore \angle ABC = 30^\circ + 50^\circ = 80^\circ$

ii) $AC = x$

$$x^2 = 30^2 + 50^2 - 2 \times 30 \times 50 \cos 80^\circ$$

$$x = 53.6568 \text{ km} \quad \checkmark$$

Question 7 (10 marks)



$$\frac{\sin \alpha}{50} = \frac{\sin 80^\circ}{53.6568} \therefore \alpha = 66^\circ 35'$$

$$\therefore \angle NAC = 150^\circ - \alpha = 83^\circ 25'$$

$$\therefore \angle NCA = 96^\circ 35' \text{ (co-int. } \angle \text{ s on } \overline{AC})$$

$$\therefore \boxed{\text{bearing}} \text{ is } 360^\circ - 96^\circ 35' = 263^\circ 25' \checkmark$$

$$b) f(x) = \frac{1}{(x-1)(x+1)} = \frac{1}{x^2-1}$$

$$f(-x) = \frac{1}{(-x-1)(-x+1)} = \frac{-1}{(x+1)(1-x)}$$

$$= \frac{-1}{1-x^2} = \frac{-1}{-(x^2-1)} = \frac{1}{x^2-1}$$

$\therefore$ even. $\checkmark$

$$c) 2x^2 - (k+1)x - 2k+3$$

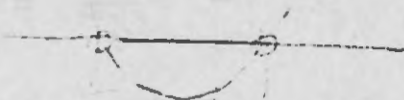
$$\Delta < 0 \text{ and } \boxed{k < 0}$$

$$b^2 - 4ac < 0$$

$$[-(k+1)]^2 - 4 \times k \times (-2k+3) < 0$$

$$k^2 + 2k + 1 + 8k^2 - 12k < 0$$

$$9k^2 - 10k + 1 < 0 \checkmark$$



$$\therefore \frac{1}{9} < k < 1 \checkmark$$

but $k < 0 \therefore$ no solution $\checkmark$

Question 8

Question 8 (13 marks)

a) since G.P.: $\frac{5x+1}{2x+2} = \frac{10x+2}{5x+1}$ ✓

$$\therefore (5x+1)^2 = (10x+2)(2x+2)$$

$$25x^2 + 10x + 1 = 20x^2 + 24x + 4$$

$$5x^2 - 14x - 3 = 0$$

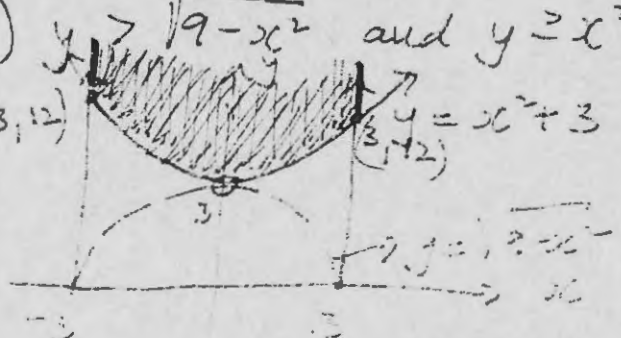
$$(5x+1)(x-3) = 0$$

$$x = -\frac{1}{5} \text{ or } x = 3 \quad \checkmark$$

b) $\sec(180^\circ - \theta) \cdot \tan(90^\circ - \theta)$
 $= \frac{1}{\cos(180^\circ - \theta)} \cdot \cot \theta \quad \checkmark$

$$= \sqrt{\frac{1}{-\cos \theta}} \cdot \frac{\cos \theta}{\sin \theta}$$

$$= \frac{1}{\sin \theta} \text{ or } -\operatorname{cosec} \theta$$

c) $y > \sqrt{9-x^2}$ and $y < x^2+3$


① semicircle

& parabola

① dotted, full boundaries

① region

d) $2x^2 - 3x + 6 = 0$

i) $\alpha + \beta = -\frac{b}{a} = \frac{3}{2} \quad \checkmark$

ii) $\alpha\beta = \frac{c}{a} = \frac{6}{2} = 3 \quad \checkmark$

iii) $\alpha^3 + \beta^3 = (\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2)$

$$= (\alpha + \beta)(\alpha^2 + \beta^2 - \alpha\beta) \quad \checkmark$$

$$= (\alpha + \beta)((\alpha + \beta)^2 - 3\alpha\beta)$$

$$= \frac{3}{2} \times \left[\left(\frac{3}{2}\right)^2 - 3 \times 3 \right]$$

$$= -\frac{81}{8} \quad \checkmark$$

Question 9 (12 marks)

a) i) $y = x^3 - 2x^2 + x$

$$\frac{dy}{dx} = 3x^2 - 4x + 1 \quad \checkmark$$

ii) $\frac{dy}{dx} = 0 = 3x^2 - 4x + 1$

$$(3x-1)(x+1) = 0$$

$$x = \frac{1}{3} \quad x = -1$$

$$y = \frac{4}{27} \quad y = 0 \quad \checkmark$$

nature:

$$\frac{d^2y}{dx^2} = 6x - 4 \quad \checkmark$$

at $x = 1 \therefore \frac{d^2y}{dx^2} = 2 > 0 \therefore (1, 0)$
 min.

at $x = \frac{1}{3} \therefore \frac{d^2y}{dx^2} = -2 < 0 \therefore \left(\frac{1}{3}, \frac{4}{27}\right)$
 max.

Question 9

a) iii) $\frac{d^2y}{dx^2} = 6x - 4$

$$0 = 6x - 4$$

$$\frac{4}{6} = x = \frac{2}{3} \quad \checkmark \quad y = \frac{2}{27}$$

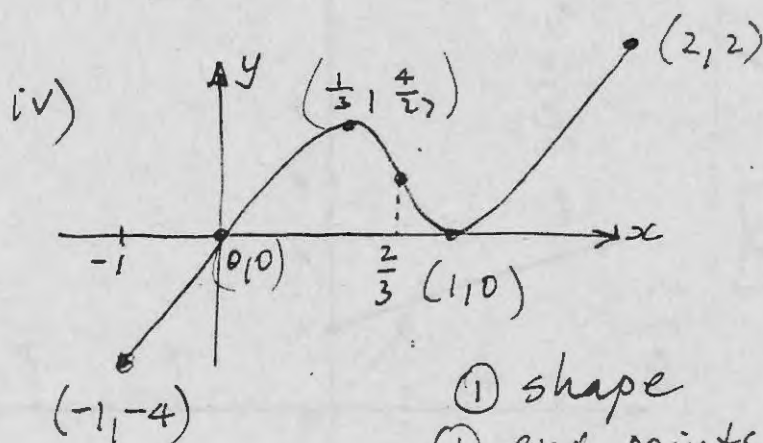
Test Concavity:-
Since at $(\frac{2}{3}, \frac{2}{27})$...

and at $x = \frac{2}{3}$ $\frac{d^2y}{dx^2} > 0 \therefore$ conc. up

$\therefore (\frac{2}{3}, \frac{4}{27})$ pt. of inf.

or

x	$\frac{1}{3}$	$\frac{2}{3}$	1
$\frac{dy}{dx}$	-2	0	$+2$



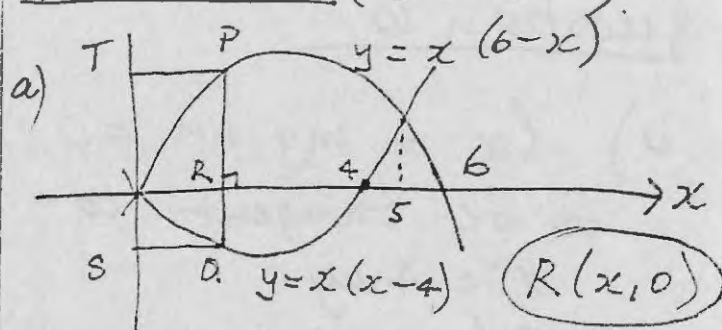
v) $\frac{2}{3} < x \leq 2$ (on domain from (iv))
or $x > \frac{2}{3}$ if all real are taken into domain

9b) LHS = $(\cot \theta + \operatorname{cosec} \theta)^2 =$

$$\cot^2 \theta + 2 \frac{\cos \theta}{\sin \theta} \cdot \frac{1}{\sin \theta} + \frac{1}{\sin^2 \theta} = \frac{\cos^2 \theta + 2 \cos \theta + 1}{\sin^2 \theta}$$

RHS = $\frac{1 + \cos \theta}{1 - \cos \theta} \cdot \frac{1 + \cos \theta}{1 + \cos \theta} = \frac{(1 + \cos \theta)^2}{1 - \cos^2 \theta} = \frac{1 + 2 \cos \theta + \cos^2 \theta}{\sin^2 \theta} = \text{LHS}$

Question 10 (8 marks)



i) $F(x, x(6-x))$ ✓

$G(x, x(x-4))$

$$\therefore PQ = \sqrt{0 + (y_P - y_Q)^2}$$

or $PQ = y_P - y_Q = x(6-x) - x(x-4)$
 $= x^2 + 4x + 6x - x^2$
 $= -2x^2 + 10x$

ii) Area = $x \times PQ$
 $= x(-2x^2 + 10x)$ ✓

$$A = -2x^3 + 10x^2$$

iii) $\frac{dA}{dx} = 20x - 6x^2$

$$0 = 2x(10 - 3x)$$

$$x = 0 \quad x = \frac{10}{3} \quad \checkmark$$

$$\frac{d^2A}{dx^2} = 20 - 12x$$

at $x = 0$ $\frac{d^2A}{dx^2} = 20 \therefore$ Min.

at $x = \frac{10}{3}$ $\frac{d^2A}{dx^2} = -20 \therefore$ Max ✓

$\therefore$ Max. A is at $x = \frac{10}{3}$

Question 10

b) $lx + my + n = 0$
to be tangent to
 $x^2 = 4ay$

$\therefore (1) \begin{cases} lx + my + n = 0 \\ x^2 = 4ay \end{cases}$ one solution

$\therefore$ simultaneously solve

(2) $y = \frac{x^2}{4a}$

(1) $lx + m \cdot \frac{x^2}{4a} + n = 0$

$x^2 \cdot \frac{m}{4a} + l \cdot x + n = 0$

for one solution $\Delta = 0$

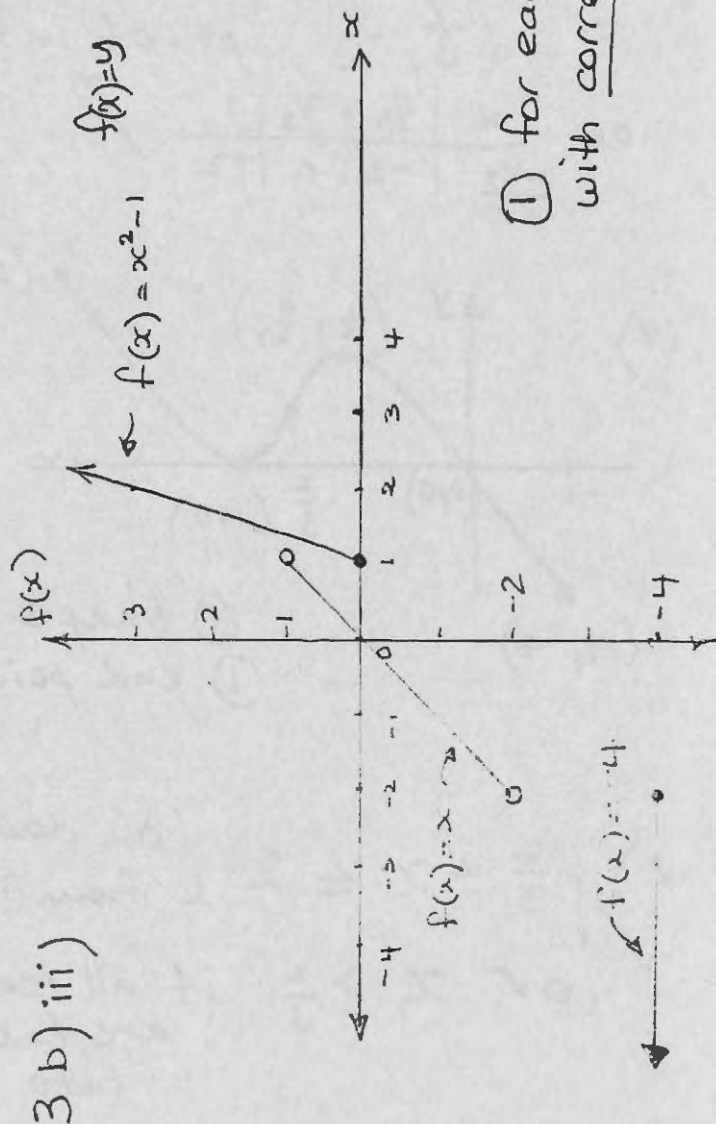
$\Delta = b^2 - 4ac$

$0 = l^2 - 4 \times \frac{m}{4a} \times n$

$l^2 = 4 \times \frac{m}{4a} \times n$

$\frac{mn}{a} = l^2$

$\therefore \frac{mn}{l^2} = a$



① for each part
with correct endpoints.